MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS)

Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBOTA08 - ARTIFICIAL INTELLIGENCE IN PLANT SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Part A
Short Answer
Answer any 10 questions
Each question carries 2 marks

1. Mention any two types of data used in AI. [K] / [CO1]
   - Images are used as visual data in AI.
   - Text is used as language data in AI.
2. How does CNN detect plant diseases? [U] / [CO2]
   - CNN detects disease by identifying spot patterns on leaves.
   - It classifies the plant condition into different disease categories.
3. What is climate resilience in agriculture? [K] / [CO3]
   - Climate resilience in agriculture refers to the ability of crops to survive stress conditions.
   - It involves adaptation to changing climate factors.
4. Define transparency in AI systems. [K] / [CO2]
   - Transparency in AI systems means providing clear outputs to users.
   - It ensures that the results are explainable and understandable.
5. Why is smart irrigation essential in precision agriculture? [U] / [CO3]
   - Smart irrigation helps in saving water through efficient usage.
   - It improves crop yield by providing the right amount of water at the right time.
6. Name one of the earliest AI programs. [K] / [CO1]
   - One of the earliest AI programs is the Logic Theorist.
   - It was designed to solve logical problems.
7. How will you collect images for disease dataset? [An] / [CO3]
   - Leaf photos are collected from infected and healthy plants.
   - The images are labelled according to disease type for training the dataset.
8. Name one technology used to collect field data. [K] / [CO2]
   - Sensors are used to collect real-time data from the field.
   - Satellite technology is used to gather large-scale field information.
9. Assess one advantage of AI used by startups. [E] / [CO3]
   - AI used by startups helps in reducing waste in agricultural inputs.
   - It increases productivity by improving efficiency in farming practices.
10. What is accountability in AI? [K] / [CO4]
    - Accountability in AI means taking responsibility for the system's decisions.
    - It ensures that outcomes are answerable to users.
11. How does a plant ID app identify plants? [U] / [CO2]
    - A plant ID app identifies plants by matching images with a database.

- It predicts the plant species based on the closest match.

12. Why is preprocessing done in image analysis? [U] / [CO2]
    - Preprocessing is done to remove noise from images.
    - It enhances image quality for better analysis.

[2x10 = 20]

Part B
Short Essay
Answer any 8 questions
Each question carries 5 marks

13. Evaluate limitations of AI agricultural platforms. [E] / [CO3]
    - AI agricultural platforms depend heavily on internet connectivity.
    - The cost of implementing AI tools can be high.
    - Data errors can affect the accuracy of predictions.
    - Farmers may face a skill gap in using advanced technologies.
    - Lack of trust can limit adoption of AI platforms.
14. Explain why data privacy is included in ICAR guidelines. [U] / [CO2]
    - Data privacy is included to protect farmers' personal and agricultural data.
    - It ensures that data is used only with proper consent.
    - It maintains security against misuse or unauthorized access.
    - It promotes ethical use of AI and digital technologies.
    - It builds trust among farmers in AI-based systems.
15. Explain the workflow of AI plant disease detection. [U] / [CO3]
    - Image data of plant leaves is collected for analysis.
    - The images are labelled according to disease type.
    - The model is trained using the labelled dataset.
    - The model is tested to check its performance.
    - The trained model predicts the disease in new images.
16. Write short notes on the role of Artificial Intelligence in agriculture. [K] / [CO1]
    - AI helps in predicting crop yield using data analysis.
    - It supports disease detection in plants at early stages.
    - AI identifies stress conditions like drought and pests.
    - It enables smart farming through automation and monitoring.
    - AI improves overall agricultural productivity.
17. What is plant stress? [K] / [CO1]
    - Plant stress refers to adverse conditions affecting plant growth.
    - It leads to reduction in normal growth and development.
    - Stress can cause damage to plant tissues.
    - It is often caused by environmental factors like drought or heat.
    - Plants respond to stress for survival.
18. Explain how AI predicts crop yield. [U] / [CO2]
    - Data is collected from sources like weather, soil and crops.
    - The model is trained using the collected data.
    - AI learns patterns from the training data.
    - The model predicts crop yield based on learned patterns.
    - The final output provides estimated yield results.
19. Suggest data governance measures for rural digital farming platforms. [A] / [CO4]

- Farmers should be made aware of consent before data collection.
- Simple and clear data policies should be provided.
- Data must be stored securely to prevent misuse.
- Access control should be implemented to limit unauthorized use.
- Training should be given to farmers on data handling practices.

20. Define unsupervised learning. [K] / [CO1]
    - Unsupervised learning is a type of machine learning in AI.
    - It works with unlabelled data without predefined outputs.
    - The system finds patterns and relationships in the data.
    - It is commonly used for clustering similar data points.
    - It helps in discovering hidden structures in datasets.
21. Define OpenCV and its importance. [K] / [CO1]
    - OpenCV is a computer vision library used in AI applications.
    - It supports image processing and analysis tasks.
    - It is used for object detection and recognition.
    - OpenCV can be implemented using Python programming.
    - It is important for developing vision-based AI systems.
22. Suggest steps to get correct plant disease output from an app. [A] / [CO4]
    - Capture a clear image of the leaf without blur.
    - Ensure good lighting while taking the photo.
    - Use a plain background to avoid distractions.
    - Upload the image correctly into the app.
    - Verify the output before taking action.

[5x8 = 40]

Part C

Essay

Answer any 1 questions

Each question carries 10 marks

23. Explain how AI supports nutrient prediction and recommendations. [U] / [CO3]
    - Data is collected from soil, crop, and environmental sources.
    - Soil testing is done to identify nutrient levels.
    - Crop stage is considered for accurate nutrient needs.
    - Weather input is included for better prediction.
    - The AI model is trained using collected data.
    - The system learns patterns related to nutrient requirements.
    - Nutrient deficiencies are detected by the model.
    - AI provides fertilizer recommendations based on deficiency.
    - It suggests the correct dose and timing of application.
    - The final output helps farmers improve nutrient management.

24. Explain why IEEE initiatives are important for global AI progress. [U] / [CO2]
    - IEEE initiatives provide common standards for AI systems.
    - They help build global trust in AI technologies.
    - They promote technology sharing across countries.
    - IEEE supports research and development in AI.
    - It encourages collaboration among scientists and developers.
    - These initiatives drive innovation in AI applications.

- IEEE emphasizes ethical use of AI systems.
- It ensures safety and reliability of AI technologies.
- It promotes wider adoption of AI solutions.
- IEEE contributes to overall development of global AI systems.

[10x1 = 10]